
Assessment of the Impact of the Circle Line Loop Closure on Multimodal Public Transport Travel Time and Path Restructuring in Singapore's Southern Region

1. Abstract

The opening of Circle Line stations CC30, CC31, and CC32 completes the orbital structure of the line, forming a closed-loop MRT network. In Singapore's dense multimodal public transport system, this structural change may alter route choices and travel times across different origin–destination pairs.

This project evaluates the impact of the Circle Line loop closure on public transport travel time from the southern region to two major destinations: the city centre and Changi Airport. A multimodal public transport network is constructed for conditions before and after the opening of the new stations. demand is represented by grid nodes connected to nearby MRT stations and bus stops. Shortest-path travel times are computed from each grid node to the destination areas.

The results show that the loop closure generates selective rather than system-wide improvement. Of 9,212 OD pairs, 4248 are restructured, and the effect is substantially stronger for CBD-bound than for airport-bound trips in both coverage and saving magnitude. Spatially, the strongest origin-side gains cluster in the central-western part of the study area, while destination-side gains accumulate mainly at selected CBD terminals such as Bayfront and Marina Bay.

AI declaration: This project used ChatGPT 5 (OpenAI) for following purpose: idea inspiration, code development, literature exploration, and writing guidance and refinement. I am responsible for the content and quality of the submitted work.

2. Introduction

Singapore's Mass Rapid Transit Circle Line is scheduled to become a complete loop in the first half of 2026 with the opening of three new stations: CC30 Keppel, CC31 Cantonment, and CC32 Prince Edward Road. According to the Land Transport Authority, these stations will connect HarbourFront and Marina Bay and close the remaining gap in the line. This change will modify the topology of the MRT network and may affect route choice and travel time.(Land Transport Authority (Singapore), 2026a)

Singapore's public transport system consists of both rail and bus networks. (Land Transport Authority (Singapore), 2026b) Accordingly, this study models both modes to represent multimodal route choice. This study asks two linked questions: whether the Circle Line loop closure improves public transport accessibility from the southern region, and through what network and routing mechanisms such improvement is produced. The project therefore evaluates the impact of the Circle Line loop closure from three perspectives:

- (1) changes in MRT network structure, with emphasis on station centrality.
- (2) origin–destination pairs whose optimal paths are reconfigured after loop closure.
- (3) changes in public transport travel time for passengers from the southern region.

3. Methodology

This study focuses on the area bounded by the West Coast Highway, Clementi Road, the Ayer Rajah Expressway, and the future Keppel station, with destinations defined within URA's Downtown Core planning area (Urban Redevelopment Authority, n.d.) and Changi Airport, and all origin–destination movements evaluated within a common network framework.

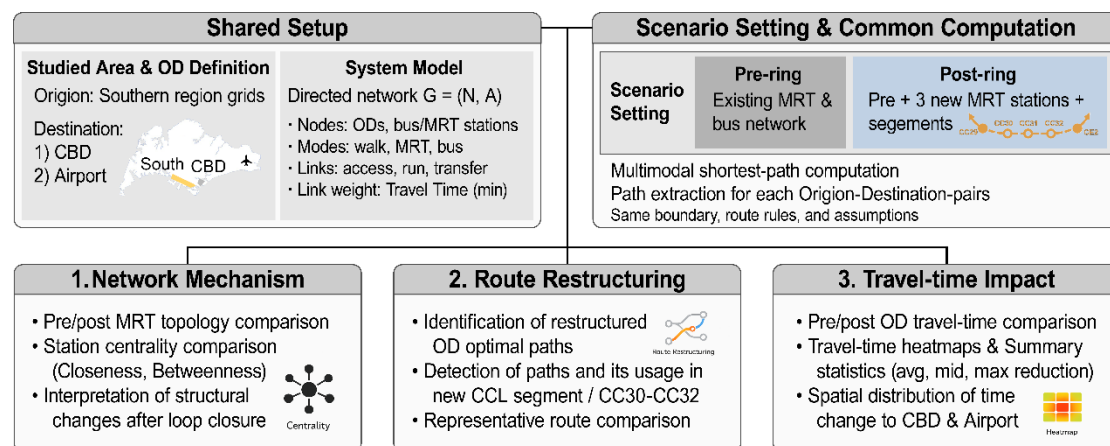


Figure 1: Methodological framework of the study

Two scenarios are constructed for comparison: a pre-ring network and a post-ring network. The only structural difference is the addition of CC line stations and the corresponding new segment $CC29 \leftrightarrow CC30 \leftrightarrow CC31 \leftrightarrow CC32 \leftrightarrow CE2$. On this basis, the methodology proceeds in three parts: network mechanism, optimal path restructuring, and travel-time impact. The first part isolates the rail-network mechanism of the loop

closure, while the latter two parts evaluate how this structural change propagates through multimodal shortest paths and travel-time outcomes in the full public transport system.

3.1. MRT Network Mechanism

MRT station centrality is evaluated separately for the pre-opening and post-opening networks to evaluate effects of the CC29↔CC30↔CC31↔CC32↔CE2 closure at the rail-network level. In this analysis, each physical MRT station is treated as a node, and adjacent-station travel time is assigned to rail links between stations, while in-station transfer is excluded. This simplified station-based representation is used to isolate the topological effect of the loop closure at the rail-network level before multimodal passenger routing is evaluated on the full graph.

Two node-level indicators from lecture notes are used to capture network mechanism. (1) Betweenness centrality B_i , measures how strongly a station participates in time-minimising movement across the network; in the results, its change is used to evaluate whether minimum-time connectivity becomes more concentrated on the new Circle Line segment and its adjoining stations after loop completion. (2) Closeness centrality C_i , measures how efficiently a station is connected to the rest of the rail network; in the results, its pre-post change is used to assess whether the loop closure improves overall network reachability around key access stations.

3.2. Optimal Path Restructuring

This section defines how minimum-travel-time OD paths are represented and extracted for comparison between the pre-ring and post-ring scenarios. Origins are defined as the centroids of 250 m grid cells, and destinations as public transport terminal nodes in the CBD and Changi Airport, thus each origin is evaluated against the same destination sets under both scenarios.

3.2.1. Intermodal Network Modelling

For routing, the transport system under scenario $s \in \{\text{pre}, \text{post}\}$ is implemented as a directed time-weighted network $G^s = (N^s, A^s)$, where N comprises (1) grid centroids, (2) bus route, (3) bus stop board, (4) bus stop alight, (5) MRT platform, (6)

MRT station hub, while A comprises (1) bus run, (2) bus board, (3) bus alight, (4) bus transfer, (5) MRT run, (6) MRT transfer, (7) intermodal transfer, (8) access. Bus and MRT services are expanded as continuous route chains and connected through station level transfer arcs, as in Figure 2.

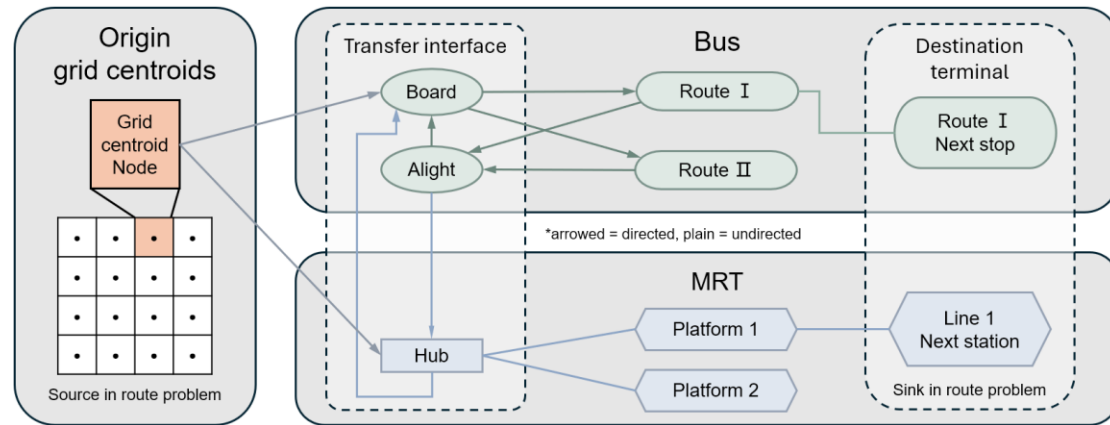


Figure 2: Station-level multimodal graph representing OD routing

Run links of bus are generated from the ordered stop sequence and adjacent-stop distance in the cleaned bus route dataset, with link travel time inferred using the representative bus operating speed. For existing MRT run links, travel times are taken from a project-compiled adjacent-station MRT link table built from raw adjacent-station observations scraped from the TransitLink Rail eGuide for the pre-ring network (TransitLink, n.d.). Where no raw observation is available, a default adjacent travel time of 2.0 minutes is applied.

Access links connect each grid centroid node to nearby bus stops and MRT stations. Access time is calculated from centroid-to-stop or centroid-to-station distance using a walking speed of 1.25 m/s (Transportation Research Board, 2003a) and a detour factor of 1.414 to approximate pedestrian detours from straight-line distance. Following the Transit Capacity and Quality of Service Manual, mode-specific access radii are set as 400 m for bus stops and 800 m for MRT stations (Transportation Research Board, 2003b), both defined as air-distance thresholds. A nearest-node fallback is further applied to ensure that every origin remains connected to both the bus and MRT layers. This fallback is introduced only to avoid disconnected origins in the routing graph and does not alter the radius-based access rule for the large majority of centroids. For bus

running links, a representative operating speed of 16 km/h is adopted, consistent with typical urban peak-hour bus performance (Federal Transit Administration, 1992).

Transfers are represented through station level node expansion. On the bus side, each physical stop is split into boarding and alighting nodes linked to route specific bus nodes, with a fixed bus transfer arc connecting the alighting node to the boarding node. On the MRT side, each physical station is represented by a hub connected to one or more line-specific platform nodes. Bus to MRT transfer with 10.3 minutes cost is encoded by directed links between bus stop alighting nodes and MRT station hubs (Yap et al., 2024). The overall structure separates boarding, alighting, same-stop bus transfer, in-station MRT transfer, and intermodal transfer within a single graph, while assigning each arc a travel-time weight under the corresponding scenario.

3.2.2. Path Comparison

This subsection defines the shortest-path model and the pre-post criteria for path restructuring. Let $s \in \{\text{pre}, \text{post}\}$ denote the network scenario, and let $o \in O$ and $d \in D$ denote an origin centroid and a destination terminal node, respectively. For each scenario, the feasible path set between o and d is denoted by Ω_{od}^s . The travel time of any feasible path $P \in \Omega_{od}^s$ is defined as

$$T^s(P) = \sum_{a \in P} t_a^s,$$

where t_a^s is the travel-time weight of arc a under scenario s . The minimum-travel-time path is then given by

$$P_{od}^{*,s} = \arg \min_{P \in \Omega_{od}^s} T^s(P),$$

which is solved on G^s by Dijkstra's algorithm. For the same OD pair, travel-time change between the two scenarios is defined as

$$\Delta T_{od} = T^{\text{post}}(P_{od}^{*,\text{post}}) - T^{\text{pre}}(P_{od}^{*,\text{pre}}).$$

To identify path restructuring, the post-ring optimal path is compared with its pre-ring counterpart using the ordered node sequence $\mathbf{n}_{od}^s$ and ordered link sequence $\mathbf{a}_{od}^s$. An OD pair is defined as path-changed when

$$I_{od}^{\text{path}} = \mathbf{1}[(\mathbf{n}_{od}^{\text{pre}} \neq \mathbf{n}_{od}^{\text{post}}) \vee (\mathbf{a}_{od}^{\text{pre}} \neq \mathbf{a}_{od}^{\text{post}})] = 1.$$

Accordingly, an OD pair is classified as restructured when its optimal path changes between the two scenarios. In addition, the use of the newly added Circle Line segment is recorded by

$$I_{od}^{\text{ring}} = \mathbf{1}[P_{od}^{*,\text{post}} \cap A_{\text{ring}} \neq \emptyset],$$

where A_{ring} denotes the set of newly added ring links. These OD-level indicators provide the basis for the path restructuring analysis in Section 4.2.

3.3. *Travel-time Impact*

Building on the minimum-travel-time formulation in Section 3.2, travel-time impact is evaluated at the origin–destination pair level by comparing the minimum travel time obtained under the pre-ring and post-ring scenarios for the same centroid–terminal node pair. Since both scenarios are solved over the same OD set and under the same minimum-travel-time routing framework, the observed time difference provides a consistent scenario-based estimate of travel-time impact associated with the network modification under the same routing assumptions.

3.3.1. Statistical Analysis

To capture both the overall extent of time savings and the extent to which such savings are concentrated in particular destination functions, the OD-level time differences are examined for the full set of trips and for destination-specific subsets, namely CBD-bound and airport-bound trips. In this way, the statistical analysis evaluates travel-time impact in terms of both coverage and magnitude, which together reflect how strongly the Circle Line closure alters time-based accessibility across the study OD system.

3.3.2. Spatial Analysis

To examine how travel-time impact is distributed across space after the Circle Line loop closure, spatial analysis is conducted from both origin and destination side of the OD system. For origins, OD-level travel-time changes are aggregated to each grid centroid for a given destination subset, resulting a surface that represents how strongly

different locations benefit, on average, from the network change. This provides the basis for heat-map comparison between CBD-bound and airport-bound trips.

For destinations, OD-level travel-time changes from all origins are aggregated to each terminal node, so that the analysis captures how strongly individual destinations are affected by the same network modification. Because destination terminals are fewer in number and spatially concentrated, especially within the CBD, their spatial effect is reported through ranked and representative terminal summaries rather than continuous surfaces. Together, these two aggregation views allow the travel-time impact to be interpreted both as a spatial pattern and as a destination-specific effect within the same OD framework.

4. Results & Discussion

This section presents the effects of the Circle Line closure from network mechanism, shortest-path restructuring, and travel-time impact. Overall, the results show that the new southern link generates selective rather than system-wide change, with its effects concentrated in specific stations, OD pairs, and destination patterns.

4.1. *MRT Station Centrality Changes*

The Circle Line closure mainly increases the structural importance of the southern corridor and its CBD-facing interface stations. The clearest gains are concentrated around Marina Bay, Raffles Place, City Hall, and HarbourFront, which suggests that the post-ring network change is spatially focused rather than system wide.

In terms of betweenness, the largest increases occur at Marina Bay, Raffles Place, City Hall, and HarbourFront. This indicates that the new segment from CC29 to CE2 takes on a larger bridge function in the MRT network, so that minimum-time station-to-station connectivity becomes more concentrated on the southern arc after loop closure. In other words, the closure changes which stations are structurally most important as intermediate connectors in the network.

Table 1: Existing stations with the largest increase in betweenness

Station	Interchange	Δ Betweenness
CE2 Marina Bay	Yes	+0.024264
EW14 Raffles Place	Yes	+0.014997
EW13 City Hall	Yes	+0.011150
CC29 HarbourFront	Yes	+0.009128

In terms of closeness, the strongest improvements are found at Marina Bay, Raffles Place, Shenton Way, Bayfront, and HarbourFront. This suggests that the loop closure improves overall network reachability most clearly around the southern CBD waterfront and its adjoining transfer interface. Among the three new stations, Prince Edward Road records the highest post-opening betweenness and closeness, indicating that it occupies the most strategic position within the newly added segment, although its network-wide importance remains moderate relative to the full post-ring system.

Table 2: Existing stations with the largest increase in closeness

Station	Interchange	Δ Closeness
CE2 Marina Bay	Yes	+0.001049
EW14 Raffles Place	Yes	+0.000962
TE19 Shenton Way	No	+0.000935
CE1 Bayfront	Yes	+0.000929
CC29 HarbourFront	Yes	+0.000925

Table 3: Newly opened stations in the post-ring network

Station	Betweenness		Closeness	
	post	rank	post	rank
CC32 Prince Edward Road	0.017241	121 / 146	0.062072	53 / 146
CC31 Cantonment	0.006561	134 / 146	0.056289	75 / 146
CC30 Keppel	0.002921	135 / 146	0.054206	80 / 146

Overall, the two indicators support the mechanism proposed in Section 3.1. Betweenness captures the redistribution of bridge function across the rail network, while closeness captures the improvement in station-level reachability to the rest of the system. Together, they suggest that the loop closure strengthens the southern

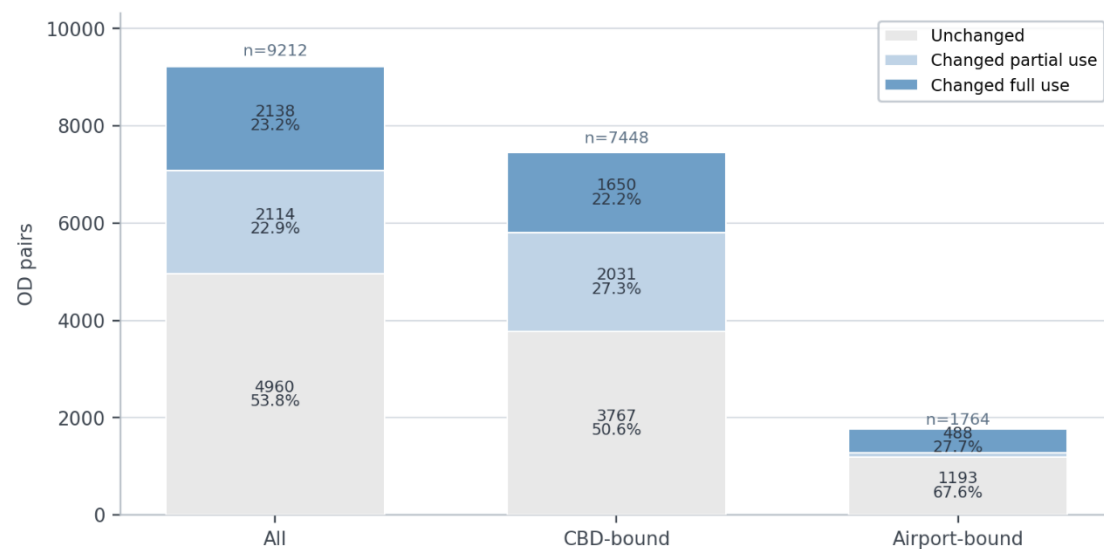
corridor and its CBD-facing interface stations within the post-ring MRT network.

4.2. *Optimal Path Restructuring*

This section compares shortest paths between the pre- and post-opening networks to identify how route structures respond to the Circle Line closure. The results show that path restructuring differs across OD pairs and mainly reflects the emergence of shorter alternatives, with only limited equivalent-time substitution.

4.2.1. Change Extent

The loop closure restructures a substantial share of minimum-travel-time OD pairs, but the effect is uneven across destination groups. Among all 9212 OD pairs, 4248 are restructured, accounting for 46.11% of the total. The restructuring share is higher for CBD-bound trips at 49.37% than for airport-bound trips at 32.37%. As shown in Figure 3 and Table 4, path change is therefore concentrated more strongly in city-centre access than in airport access.



Note: Path change without use of the new segment was not observed.

Figure 3: Path change and new-segment use by destination group

Table 4: Path restructuring by destination group

Destination group	Total OD pairs	Restructured OD pairs	Share (%)
All	9212	4248	46.11
CBD-bound	7448	3677	49.37
Airport-bound	1764	571	32.37

4.2.2. Segment Utilization

All restructured OD pairs use at least one of the newly added Circle Line run segments. As shown in Table 6, no path change is observed without involvement of the new segment. The new segment is used in two forms. Overall, 2,134 restructured OD pairs, or 50.24%, traverse the full added segment between CC29 and CE2, while 2,114, or 49.76%, use only part of it. This indicates that the new segment functions both as a through corridor and as a partial access corridor, rather than only as an end-to-end bypass.

Table 6: New-segment use among restructured OD pairs

Destination group	Restructured OD pairs	New seg. count	Share (%)
All	4248	4248	100.00
CBD-bound	3677	3677	100.00
Airport-bound	571	571	100.00

The redistribution of path counts across MRT lines and bus services is reported in Figure 4(a). The scale of change is larger for MRT than for bus, indicating that the dominant routing effect is rail-based, while bus services adjust mainly at the feeder or terminal-access level. The largest increases occur on the Circle Line Extension and the Circle Line, whereas the largest decreases occur on the North-East Line, East-West Line, and Downtown Line. Bus services also change, but their magnitudes are smaller.

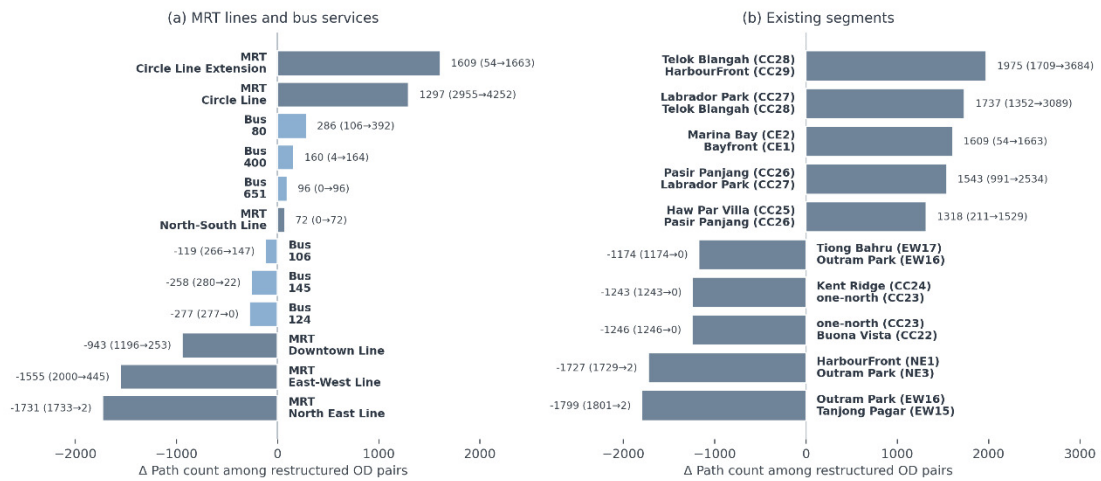


Figure 4: Path-count redistribution after loop closure

This pattern is reinforced at the segment level. As shown in Figure 4(b), path-count increases are concentrated along the western Circle Line corridor, from Haw Par Villa (CC25) to HarbourFront (CC29), and further along the extension from Marina Bay (CE2) to Bayfront (CE1). By contrast, the largest decreases occur on former CBD-facing approaches, especially from Kent Ridge (CC24) and one-north (CC23) toward Buona Vista (CC22), and from HarbourFront (NE1), Tiong Bahru (EW17), and Outram Park (EW16) toward the city centre. This pattern indicates a reorganisation of modelled minimum-time rail access from the southern origins to the CBD.

4.3. Travel-time Impact

This section presents the travel-time impact of the Circle Line closure at the system, destination-group, and spatial levels. Overall, the results indicate that time savings are selective rather than uniform, with clearer benefits concentrated in particular destination groups and locations.

4.3.1. Statistical Summary

Building on the path restructuring results in Section 4.2, the statistical analysis examines whether the Circle Line closure translates into consistent travel-time savings at the origin–destination level. Because the same OD set is evaluated under the same minimum-travel-time framework in both scenarios, the pre-ring minus post-ring difference provides a consistent measure of modelled travel-time saving. Overall, the results indicate that travel-time savings are evident but uneven, with a clearer effect

for CBD-bound trips than for airport-bound trips.

Table 5: Summary statistics of OD-level travel-time saving by destination group

Metric	Overall	CBD-bound	Airport-bound
Coverage			
OD pairs (N)	9212	7448	1764
Improved OD pairs, n (%)	4248 (46.1%)	3677 (49.4%)	571 (32.4%)
Unchanged OD pairs, n (%)	4964 (53.9%)	3771 (50.6%)	1193 (67.6%)
Magnitude of saving among all OD pairs			
Mean saving (min)	1.835	2.155	0.485
Median saving (min)	0	0	0
90th percentile saving (min)	6.772	7.144	1.4
Maximum saving (min)	16.351	16.351	7.761
Magnitude of saving among improved OD pairs			
Mean saving (min)	3.98	4.365	1.499
Median saving (min)	3.4	3.909	1.4
90th percentile saving (min)	7.8	7.8	2.102

As summarised in Table 7, the loop closure improves travel time for 4,248 OD pairs and leaves the remaining 4,964 unchanged. The resulting benefit is therefore selective in coverage, magnitude, destination group, and spatial distribution. CBD-bound trips show both wider coverage and larger typical savings than airport-bound trips.

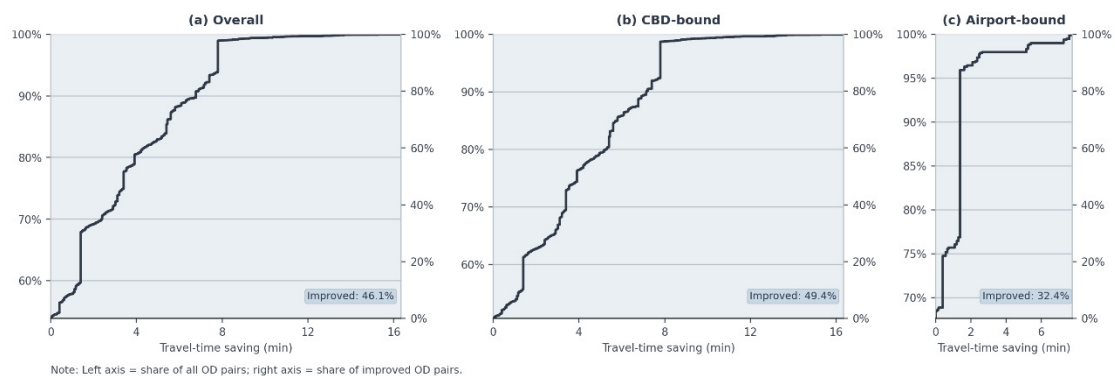


Figure 5: Cumulative distribution of travel-time saving by destination group

Figure 5 further shows that the distribution of travel-time saving differs systematically by destination group. The overall curve confirms that positive savings affect a substantial subset of OD pairs, but the contrast between the CBD-bound and airport-bound subsets is more informative. Savings for CBD-bound trips are observed for a larger share of OD pairs and extend across a wider range of magnitudes, whereas airport-bound savings are more limited in coverage and more concentrated within a

narrower range. This suggests that the loop closure has a stronger modelled travel-time effect on city-centre access than on airport access, even though both destination groups benefit from the same network change.

Overall, the OD-level statistical results are consistent with the mechanism established in Sections 4.1 and 4.2. In the updated results, all restructured OD pairs record positive travel-time saving, but the strength of that benefit depends on destination group. This statistical difference provides the basis for the next subsection, which examines how these savings are distributed across origins in the southern study area.

4.3.2. Origin Side Spatial Analysis

Origin-side aggregation shows that travel-time benefit is spatially uneven across the southern region and clearly stronger for CBD-bound than for airport-bound trips. Figure 6 maps the mean origin-level travel-time saving obtained by averaging OD-level savings from each centroid to all terminals within a given destination subset. The overall pattern indicates that the loop closure generates concentrated gains in selected zones rather than a uniform improvement across the study area.

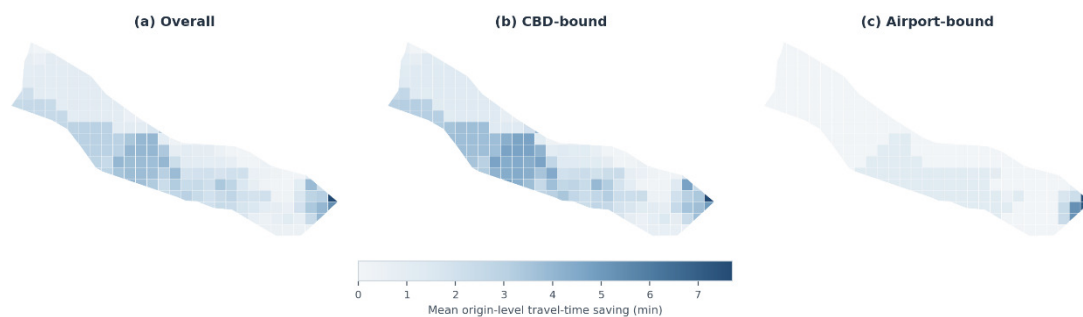


Figure 6: Heatmap of mean origin-level travel-time saving by destination subset

The overall map shows a clear but moderate pattern, with higher mean savings concentrated in a limited part of the study area rather than distributed evenly across all origins. As shown in Figure 6(a), most origins record positive average saving, but the strongest gains cluster in the central-western portion of the southern region, while other areas experience smaller improvements. This indicates that the network change benefits a substantial share of origins, but that the intensity of that benefit remains spatially differentiated.

The CBD-bound map shows that this spatial differentiation is driven mainly by city-centre access. In Figure 6(b), the highest origin-level savings are more extensive and more continuous than in the overall pattern, with a pronounced concentration in the western and central parts of the study area. This suggests that the new southern Circle Line segment strengthens access to CBD terminals for a relatively wide set of origins, and that the main origin-side gains identified in the overall map are primarily associated with CBD-bound trips.

The airport-bound map shows a weaker and more selective spatial effect. In Figure 6(c), average saving remains positive for some origins, but the magnitude is much smaller and the high-value area is more localised than in the CBD-bound case. The airport-side benefit is therefore limited to a narrower subset of locations, which is consistent with the lower OD-level coverage and smaller saving magnitude reported in Section 4.3.1. Overall, origin-side spatial analysis indicates that the loop closure produces uneven travel-time benefits and that these benefits are stronger for CBD-bound travel, which motivates the destination-side analysis in the next subsection.

4.3.3. Destination Side Spatial Analysis

Destination-side aggregation shows that terminal-level effects are uneven and stronger for CBD-bound trips than for airport-bound trips. As shown in Figure 7, several CBD terminals record both large mean savings and clearly positive medians, whereas airport terminals show only small mean savings and zero medians throughout. The loop closure therefore changes destination-side travel-time outcomes more clearly for city-centre terminals than for airport terminals.

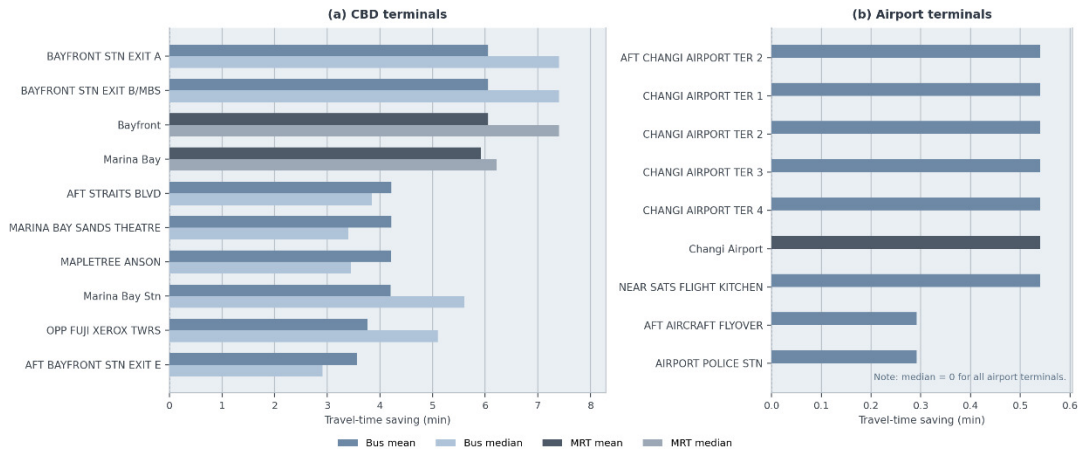


Figure 7: Terminal-level mean and median travel-time saving from all origins

Within the CBD, the benefit is concentrated rather than uniform. The largest gains occur at Bayfront, Marina Bay, and nearby bus terminals, where mean and median savings are both high, indicating that the improvement is not driven only by a small subset of origins. By contrast, other CBD terminals improve with smaller magnitudes. The destination-side effect is thus selective within the CBD, with the new southern segment strengthening modelled travel-time access most clearly to the southern waterfront and adjacent city-centre terminals.

For airport terminals, the effect is weaker and more selective. Although mean saving remains positive for most airport terminals, the median is zero for all of them, which indicates that improvement is observed for only part of the origin set rather than for most origins serving each terminal. The airport-side pattern is therefore one of limited average gain without broad-based improvement, which is consistent with the lower coverage and smaller saving magnitude already observed in the OD-level statistics.

Overall, destination-side analysis identifies where the time savings ultimately accumulate in the terminal system. Together with the origin-side heat maps in Section 4.3.2, it shows that the loop closure concentrates the strongest travel-time benefits around selected CBD terminals, while airport effects remain comparatively modest. This completes the spatial interpretation of travel-time impact from both sides of the OD system.

5. Conclusion

This study provides a scenario-based assessment of the Circle Line loop closure within a common multimodal public transport network. Using pre-ring and post-ring scenarios, it evaluates travel from the southern region to the CBD and Changi Airport through three linked components: MRT network mechanism, optimal path restructuring, and travel-time impact under the same minimum-travel-time framework. Taken together, the analysis establishes a consistent basis for comparing how a single rail-network modification propagates through network structure, modelled minimum-time paths, and OD-level travel time.

The main finding is that the loop closure produces selective rather than uniform improvement. At the rail-network level, the clearest structural gains are concentrated around Marina Bay, Raffles Place, City Hall, HarbourFront, and the new southern segment, indicating a stronger southern corridor and CBD-facing interface. At the routing level, 4,248 of 9,212 OD pairs are restructured, and all restructured pairs record lower modelled minimum travel time in the updated results; at the travel-time level, the effect is stronger for CBD-bound trips than for airport-bound trips in both coverage and magnitude. Spatially, the strongest origin-side gains cluster in the central-western part of the study area, while destination-side gains accumulate mainly at selected CBD terminals such as Bayfront and Marina Bay. Overall, the closure improves modelled travel-time access in a spatially and functionally differentiated way.

The main contribution of the study is that it connects network mechanism, path restructuring, and travel-time impact within one interpretable framework. Rather than reporting only centrality change or only travel-time saving, the analysis shows how a rail-network structural change is associated with rail-based path reorganisation and then translated into measurable OD-level and spatially aggregated time benefit. By combining origin-side and destination-side aggregation, the study also identifies where the benefit is generated and where it ultimately accumulates. Overall, the findings suggest that the main effect of the Circle Line closure lies in strengthening CBD-oriented travel-time access from the southern region, while airport-oriented improvement remains more limited.

From a planning perspective, the results suggest that the main benefit of the loop closure lies in strengthening CBD-oriented travel-time access from the southern region, while airport-oriented improvement remains more limited. This implies that further improvement in airport access may require additional network or service interventions beyond the Circle Line closure itself.

The study also has clear scope limits, and these define the main gaps for further work. The analysis is based on a minimum-travel-time shortest-path framework with fixed parameter representations for bus speed, access, MRT run time, and transfer, so it does not incorporate schedule waiting, crowding, fare, or behavioural heterogeneity in route choice. The travel times of future MRT links are represented using available adjacent-link information and default assumptions where observations are unavailable, so the estimates should be revisited when official operating times are released. In addition, although the network change may also affect transfer burden, this study does not separately quantify transfer counts or transfer-type changes, and the analysis is limited to the southern study area and two destination groups. Future work may therefore extend the framework toward schedule-based or demand-weighted modelling and examine a wider set of trip purposes and destination systems.

6. Reference

Federal Transit Administration, 1992. Characteristics of Urban Transportation Systems, Revised Edition. Washington, DC.

Land Transport Authority (Singapore), 2026a. Circle Line 6.

Land Transport Authority (Singapore), 2026b. Public Transport System.

TransitLink, n.d. TransitLink eGuide: Rail Travel Information.

Transportation Research Board, 2003a. Transit Capacity and Quality of Service Manual, Part 7: Stop, Station, and Terminal Capacity, in: Transit Capacity and Quality of Service Manual, TCRP Report 100. Transportation Research Board, Washington, DC.

Transportation Research Board, 2003b. Transit Capacity and Quality of Service Manual, Part 3: Quality of Service, in: Transit Capacity and Quality of S

ervice Manual, TCRP Report 100. Transportation Research Board, Washington, DC.

Urban Redevelopment Authority, n.d. URA SPACE: Urban Design Guidelines for Downtown Core.

Yap, M., Wong, H., Cats, O., 2024. Passenger valuation of interchanges in urban public transport. J. Public Transp. 26, 100089. <https://doi.org/10.1016/j.jpubtr.2024.100089>